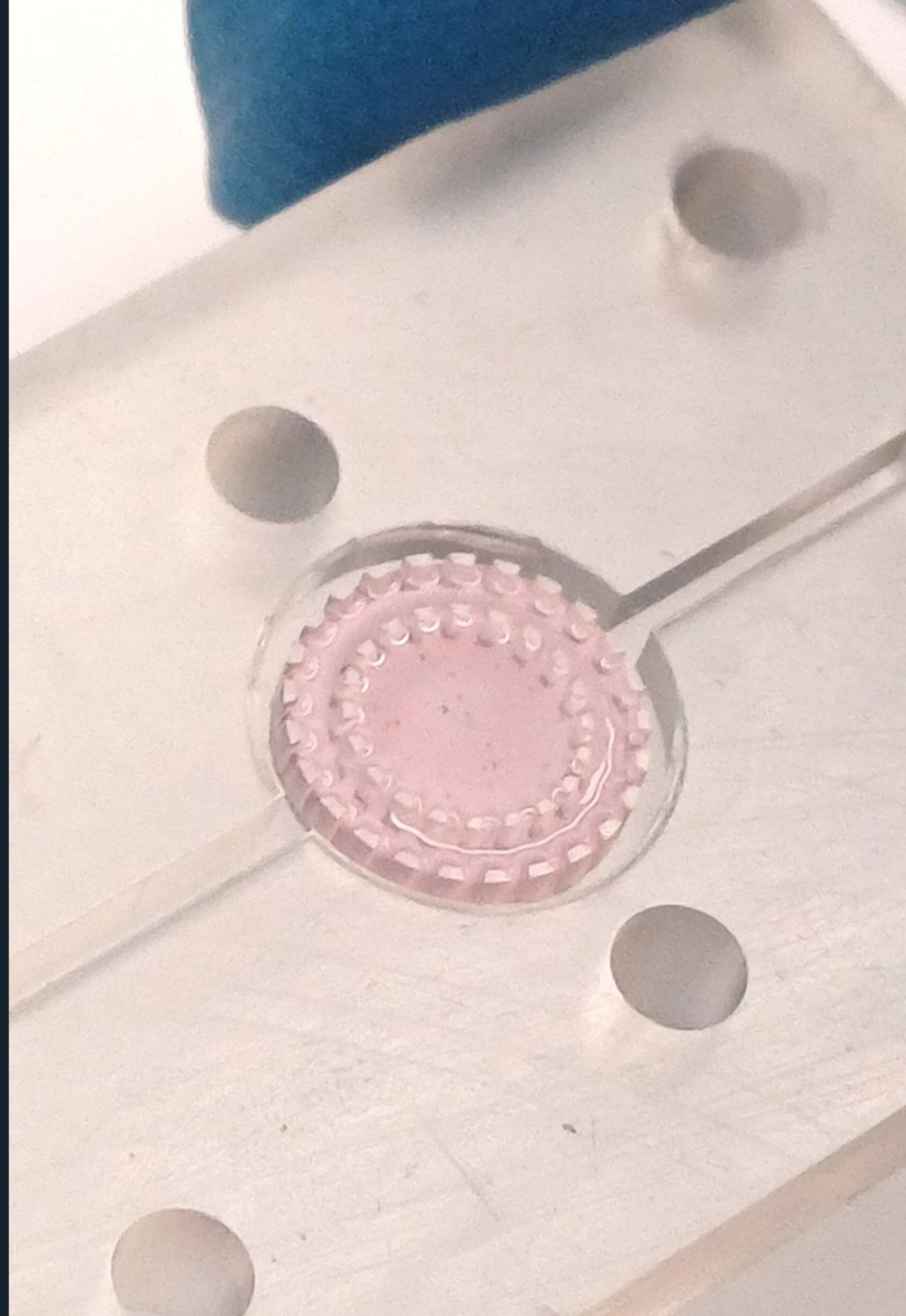
CSMI MASTER 2 · SEMESTER 3 PROJECT · 2026–2027

Calibrating tumour–astrocyte migration models on registered organ-on-chip data

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Host organisation and partners

HOST

CNR NANOTEC, Lecce

Institute of Nanotechnology, Italian National Research Council. We design 3D-printed organ-on-chips, run the imaging experiments and model transport in them.

PARTNER

Cemosis – IRMA, Unistra

Feel++ and its CFPDE toolbox for PDE simulation; home of the CSMI programme and of the previous M1 projects.

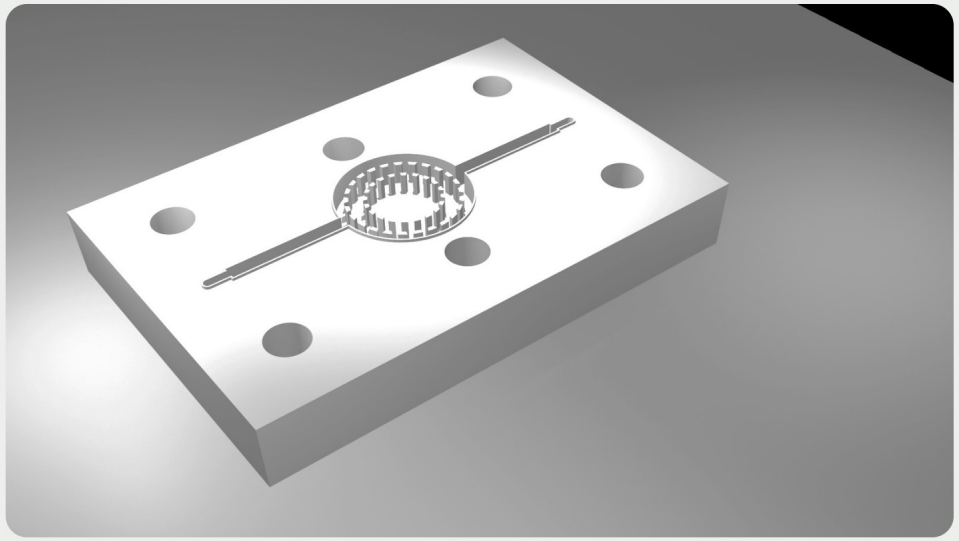
PARTNER

CNR-IAC, Rome

Mathematical modelling of tumour–astrocyte interactions.

Framework: D3 4Health (Italian PNRR), digital and biological twins for healthcare. Shared goal: turn organ-on-chip experiments into calibrated, predictive models.

The system: a brain organ-on-chip



CELLS	U87 glioblastoma (GFP), inner chamber; SVGA astrocytes (Tomato), outer chamber
MATRIX	Matrigel, perfused at 100 µL/h
IMAGING	Confocal time-lapse, 63 h at 1 frame/h; z-stack of 250 µm (25 × 10 µm)
CHIP	DLP 3D-printed, tape-sealed; fiducial tags visible for registration

CHIP 1 · 2 SCENES

Control: U87 cells only

CHIP 2 · 2 SCENES

Co-culture, untreated

CHIP 3 · 3 SCENES

Co-culture, U87 pre-treated with Ala-2/5

Starting point: two M1 projects (2025–26)

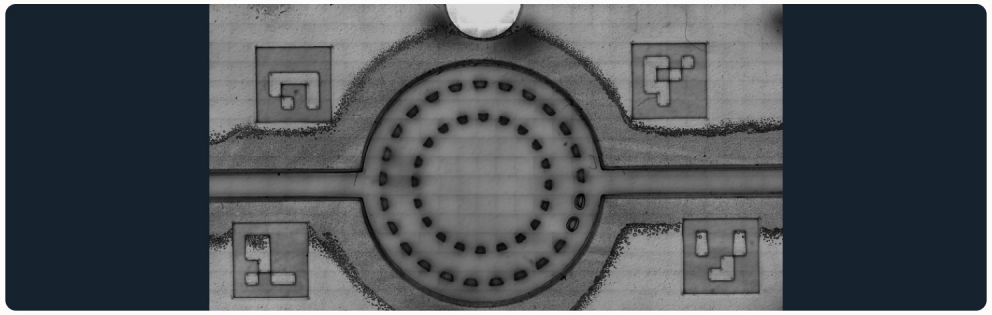
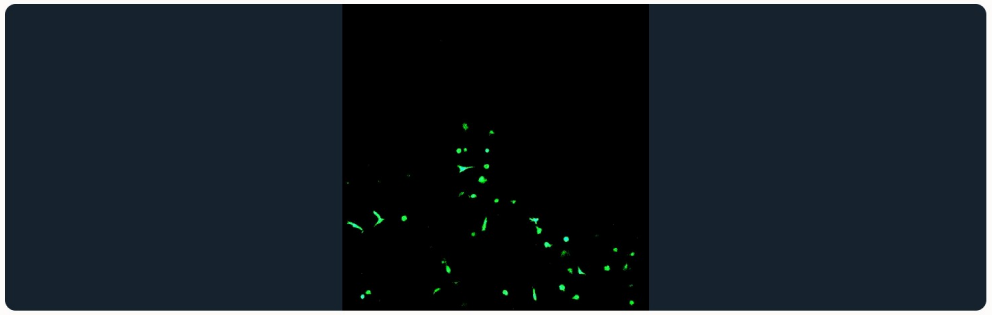


Image registration

Fiducial tags map each field of view onto the chip CAD frame (affine / projective).



Cell tracking and density maps

Trajectories and KDE density maps of tumour cells and astrocytes over time.

Preliminary result: Ala-2/5 seems to reverse the net direction of tumour–astrocyte migration, but the effect is small compared with random motility and rests on few cells.

THE PROJECT IN ONE QUESTION

Does Ala-2/5 change the sign or the strength of tumour–astrocyte cross-taxis?

Aim: estimate an effective taxis coefficient χ, with confidence intervals, and compare it across control, co-culture and co-culture + Ala-2/5.

A reduced diffusion–taxis model

$$\partial T/\partial t = \nabla\cdot(D_T \nabla T - \chi\, T \nabla A)$$

$$\partial A/\partial t = \nabla\cdot(D_A \nabla A)$$

T, A: densities of U87 cells and astrocytes, solved on the chip geometry with Feel++ CFPDE (HDG / finite elements).

$\chi > 0$: tumour cells drift toward the astrocytes; $\chi < 0$: away from them.

Two populations only. Chemical signals are dropped: imaging data cannot identify them.

Oxygen as background. A fixed field from our validated O_2 model of the chip.

Three unknowns. D_T, D_A and χ, with one χ per condition.

Work plan in three phases

PHASE 1 · WEEKS 1–3

Consolidate

- Merge registration and tracking into one CLI
- One trajectory schema in chip coordinates
- Run on the 7 new scenes, with registration error in µm

PHASE 2 · WEEKS 4–6

Measure

- MSD, persistence and directed displacement per population
- KDE density maps: from cells to continuum fields
- Non-parametric tests; sensitivity to segmentation and linking

PHASE 3 · WEEKS 7–12

Calibrate

- Twin experiment on synthetic data: what is identifiable?
- D_T and D_A from controls and MSD
- χ per condition: profile likelihood or Bayesian

Mathematical and computing work

Mathematics

- Kernel density estimation: cells → density fields
- Trajectory statistics: MSD, persistence, non-parametric tests
- Diffusion–taxis PDEs with HDG / finite elements
- Inverse problems: least squares, profile likelihood, Bayesian inference
- Sensitivity analysis, identifiability, uncertainty quantification

Computing

- Python: NumPy, SciPy, pandas, scikit-image, OpenCV
- CZI readers for the confocal data
- Feel++ CFPDE toolbox and its Python interface
- Gmsh and ParaView for meshing and visualisation
- SciPy optimize, emcee or PyMC for calibration
- Git/GitHub, Jupyter and a packaged CLI

Deliverables and success levels

ROI registration → cell trajectories → migration observables → calibrated PDE

LEVEL 1

Minimum

Reproducible pipeline on the new dataset, with observables for all three conditions.

LEVEL 2

Target

Twin experiment, then D_T and D_A calibrated with confidence intervals.

LEVEL 3

Full

χ with and without Ala-2/5, with uncertainty and a statement of what is identifiable.

Delivered as a documented Git repository (CLI and notebooks), a Feel++ case on the chip geometry, a written report and a final presentation.

Profile, supervision and follow-on

STUDENT PROFILE

Python and scientific computing, PDEs and finite elements, statistics. Curiosity for biology; no wet-lab work. Feel++ experience is a plus.

SUPERVISION AND DATA

Alessandro De Giorgi (CNR NANOTEC), with Cemosis. Data ready: 3 chips, 7 scenes, 63 h each, plus the chip CAD.

FOLLOW-ON

Internship: yes, extending the model and the analysis. PhD: possible, depending on results and funding.

Thank you · alessandro.degiorgi@unisalento.it